4. Using the provided Coffee Shop Sales dataset create an interactive Coffee Shop Sales Dashboard in Power BI Desktop with these requirements.

a) **Import Excel data sheet 'Coffee Shop Sales Data'; check/fix data types in Power Query if needed, then load to model and Close & Apply.**
b) **Set canvas background: import provided coffee shop image from description, fit to page for attractive professional dashboard appearance.**
c) **Add main title textbox 'Coffee Shop Sales Dashboard' (font size 28, white text, centered, bold with shadow or background effect).**
d) **Create 4 formatted KPI Cards with titles enabled, white borders/shadows, colored backgrounds: Total Transactions (COUNTROWS Transaction ID), Total Quantity (SUM Quantity), Total Revenue (SUM Revenue), Total Profit (SUM Profit).**
e) **Add 3 interactive slicers with titles, white borders/shadows, dropdown style: Year (from Date), Month (from Date), Shop Name for dynamic filtering across visuals. Ensure slicers filter all visuals/KPIs correctly; build visuals with white data labels, borders, shadows, black axes: Line/Clustered Column (Revenue & Quantity by Branch), Donut/Pie (Quantity by Shop Name), Column (Revenue by City), Treemap (Quantity by Product Category), Donut (Revenue by Product Category), small card (Average Quantity).**

Aim

The primary aim of the question is to build an attractive, interactive **Coffee Shop Sales Dashboard** in Power BI Desktop using the provided "Coffee Shop Sales Data" Excel sheet. It emphasizes creating a visually appealing report with a thematic background (coffee shop image), professional title, key performance indicator (KPI) cards for core sales metrics, time- and location-based slicers for filtering, and a mix of chart types to analyze revenue, quantity, and performance across branches/shops, cities, product categories, and time periods. The dashboard should demonstrate strong Power BI skills in data loading, formatting, interactivity, and thematic design, while being saved as a .pbix file and prepared for viva-style explanation of insights (e.g., top-performing shops or categories) or challenges (e.g., date handling or background fitting).

Steps to Solve the Question

Follow these step-by-step instructions to create the dashboard as specified. The dataset is likely a typical coffee shop sales Excel file (common on Kaggle/GitHub) with columns such as: Transaction ID, Date (transaction_date), Quantity (transaction_qty), Revenue (often calculated as unit_price × quantity), Profit (if pre-calculated or derivable), Shop Name / Store Name / Branch (store_location or store_id mapped to name), City (derived from location), Product Category (product_category), etc. If Revenue or Profit aren't direct columns, create them via DAX or Power Query using unit_price if available.

Step 1: Import and Prepare Data (Part a)

1. Open Power BI Desktop → **Get Data** → **Excel** → Select the file → Choose the sheet named **Coffee Shop Sales Data** → Load or **Transform Data**.
2. In Power Query Editor:
    - Check/fix data types:

        - Date column → Change Type → Date.
        - Quantity → Whole Number.
        - Revenue, Profit (if present) → Decimal Number or Currency.
        - Other columns (Shop Name, Product Category, City) → Text.
    - If needed, add columns:
        - Year = YEAR([Date])
        - Month = FORMAT([Date], "MMMM") or MONTH([Date]) for name/number.
    - Remove any completely blank rows or fix obvious errors (e.g., nulls in key fields).
3. Click **Close & Apply** to load the data model.

Step 2: Set Canvas Background (Part b)

1. On the report page → **Format** pane (paint roller icon) → **Canvas background**.
2. Click **Add image** or browse → Import a coffee shop-themed image.
    - Suggested sources (free for use): Search online for "cozy coffee shop interior background" on sites like:
        - Freepik.com (coffee background vectors/photos)
        - Pexels.com (free coffee shop background stock photos)
        - Unsplash.com (high-quality coffee shop/cafe images)
    - Choose a warm, inviting image (e.g., coffee cups on table, cafe interior, beans, or subtle brown tones) — avoid busy/overly detailed ones.
3. Set **Fit** to **Page** (or **Actual size** if needed) → Adjust **Transparency** to 20–40% for readability of visuals over the image → Apply for an attractive, professional look.

Step 3: Add Main Title (Part c)

1. **Insert** → **Text Box**.
2. Type: **Coffee Shop Sales Dashboard**.
3. Format:
    - Font size: 28
    - Color: White
    - Bold: Yes
    - Alignment: Center
    - Add shadow (Format → Effects → Shadow) or semi-transparent dark background rectangle behind it for contrast.
4. Position at the top center.

Step 4: Create 4 KPI Cards (Part d)

1. Create or use these measures (Modeling tab → New measure):
    - **Total Transactions** = COUNTROWS('Coffee Shop Sales Data') (or DISTINCTCOUNT([Transaction ID]))
    - **Total Quantity** = SUM([Quantity])
    - **Total Revenue** = SUM([Revenue])
    - **Total Profit** = SUM([Profit])
2. Drag each measure to a separate **Card** visual.
3. Format each Card:
    - Enable Title (e.g., "Total Transactions", "Total Revenue").
    - White borders/shadows (Format → General → Effects → Border & Shadow).
    - Colored backgrounds (e.g., soft brown/green/teal via Format → Background).
    - Currency formatting for Revenue/Profit, whole numbers for others.

4. Arrange in a row below the title.

Step 5: Add 3 Interactive Slicers (Part e)

1. Create slicers:
    - **Year**: Drag Year (from Date) → Dropdown style.
    - **Month**: Drag Month (from Date) → Dropdown.
    - **Shop Name**: Drag Shop Name (or store_location) → Dropdown.
2. For each slicer:
    - Add title (e.g., "Select Year").
    - White borders/shadows (Format → General → Effects).
    - Dropdown style for clean look.
3. Position them (e.g., left sidebar or top row) → They will automatically filter all visuals and KPIs (verify by selecting a value).

Step 6: Build Visuals (Part e)

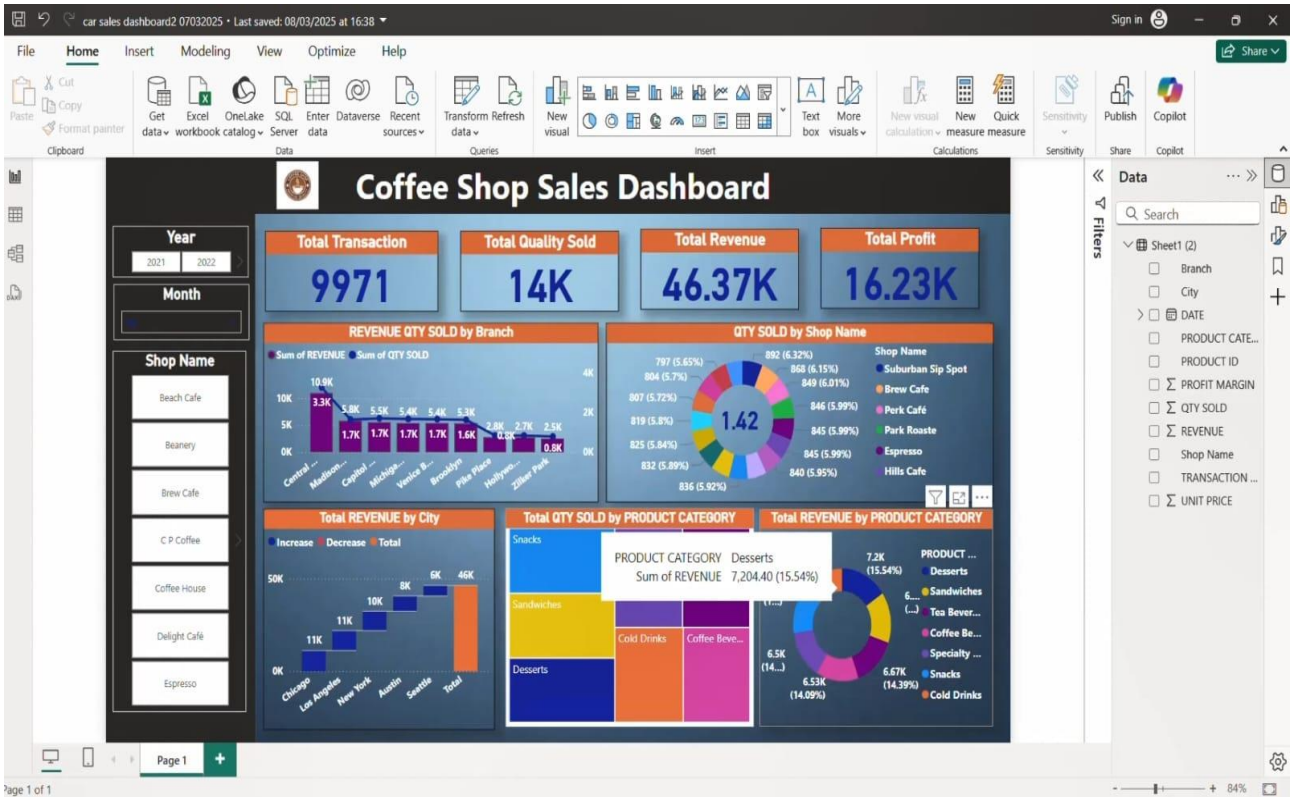
Apply consistent formatting: White data labels, borders, shadows (Format → General → Effects); black axes/labels for contrast on the background.

1. **Line/Clustered Column (Revenue & Quantity by Branch)**:
    - Clustered Column Chart or Combo Chart → Axis: Branch/Shop Name → Values: Total Revenue (column) + Total Quantity (line).
2. **Donut/Pie (Quantity by Shop Name)**:
    - Donut Chart → Legend: Shop Name → Values: Total Quantity.
3. **Column (Revenue by City)**:
    - Column Chart → Axis: City → Values: Total Revenue.
4. **Treemap (Quantity by Product Category)**:
    - Treemap → Group: Product Category → Values: Total Quantity.
5. **Donut (Revenue by Product Category)**:
    - Donut Chart → Legend: Product Category → Values: Total Revenue.
6. **Small Card (Average Quantity)**:
    - Card visual → Measure: AVERAGE([Quantity]) → Title: "Average Quantity per Transaction".
    - Format similarly (white borders/shadows, colored background).

Arrange visuals logically (e.g., KPIs top, slicers left, main charts center/bottom).

Step 7: Final Checks, Save, and Prepare for Viva

- Test: Select a Month/Year/Shop Name → Confirm KPIs, charts, and filters update (e.g., one shop shows higher revenue in a specific category).
- Ensure theme consistency (white text/labels readable on background).
- Save: File → Save As → "CoffeeShopSalesDashboard.pbix".
- **Insights examples**:
    - "Top shop/branch drives 40% of total revenue; Coffee category dominates quantity sold; Peak months show seasonal trends."
- **Challenges examples**:
    - Fitting background image without obscuring text (adjust transparency); Deriving Year/Month if not present; Ensuring slicers sync correctly with all visuals.


car sales dashboard2 07032025 • Last saved: 08/03/2025 at 16:38
File
Home
Insert
Modeling
View
Optimize
Help
Coffee Shop Sales Dashboard
Year
2021
2022
Month
Shop Name
Beach Cafe
Beanery
Brew Cafe
C P Coffee
Coffee House
Delight Café
Espresso
Total Transaction
9971
Total Quality Sold
14K
Total Revenue
46.37K
Total Profit
16.23K
REVENUE QTY SOLD by Branch
Sum of REVENUE
Sum of QTY SOLD
QTY SOLD by Shop Name
Shop Name
Suburban Sip Spot
Brew Cafe
Perk Café
Park Roaste
Espresso
Hills Cafe
1.42
Total REVENUE by City
Increase
Decrease
Total
Total QTY SOLD by PRODUCT CATEGORY
Snacks
Sandwiches
Desserts
Cold Drinks
Coffee Beve...
PRODUCT CATEGORY Desserts
Sum of REVENUE 7,204.40 (15.54%)
Total REVENUE by PRODUCT CATEGORY
PRODUCT ...
Desserts
Sandwiches
Tea Bever...
Coffee Be...
Specialty ...
Snacks
Cold Drinks
Data
Search
Sheet1 (2)
Branch
City
DATE
PRODUCT CATE...
PRODUCT ID
PROFIT MARGIN
QTY SOLD
REVENUE
Shop Name
TRANSACTION ...
UNIT PRICE
Filters
Page 1
Page 1 of 1
84%